



Lecture 2: EM Wave Propagation Fundamentals

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Maxwell's Equations

Maxwell's theory of electrodynamics describes EM fields in terms of space and time variations of EM components, allowing us to understand the formation, emission, and propagation of EW wave. There are **5 vector fields and 1 scalar field** (in **c.g.s units**):

- ***E***: Electric field intensity (statV cm⁻¹ or dyn esu⁻¹)
- ***D***: Electric displacement (esu cm⁻²)
- ***H***: Magnetic field intensity (Oe or oersted)
- ***B***: Magnetic induction or Magnetic flux density or Magnetic field (G or Gauss)
- ***J***: Electric current density (esu s⁻¹ cm⁻²)
- **ρ** : Electric Charge density (esu cm⁻³) (For free charge not for bound charge as in polarization)

Comparing with **SI units**:

- 1 esu = 1 statC = 3.33564×10^{-10} C
- 1 statV = 299.792458 V = $\frac{c}{10^8 \text{ cm/s}} \text{ V}$
- 1 statV $\times$ 1 statC = 1 erg = 1 dyne cm
- 1 Oe = $1000/(4\pi)$ A/m = 79.577 A/m
- 1 G (has a unit of dyne/esu) = 10^{-4} T

Here dyne = g cm/s², esu = cm $\sqrt{\text{dyne}}$, statV = $\sqrt{\text{dyne}}$, esu = statV cm. And ***E***, ***D***, ***B***, ***H*** can all convert to the same unit.

Maxwell's Equations - Material Eqs

Some of the field components are related by the properties of the medium in which they exist. These are the so-called material Equations:

Ohm's law:

$$\mathbf{J} = \sigma \mathbf{E}$$

and

$$\mathbf{D} = \epsilon \mathbf{E} = \mathbf{E} + 4\pi \mathbf{P}$$

$$\mathbf{B} = \mu \mathbf{H} = \mathbf{H} + 4\pi \mathbf{M}$$

where

- σ : specific conductivity (esu/s/cm/statV = 1/s)
- ϵ : permittivity or dielectric constant (unitless)
- μ : magnetic permeability (unitless)

** In vacuum, $\epsilon = 1$ and $\mu = 1$. ($\therefore$ measured w.r.t. the free space)

$\mathbf{D}$ also known as electric flux density, the charge per unit area that would be displaced across a layer of conductor placed across an electric field. $\mathbf{P}$: polarization density of any magnetic material. $\mathbf{M}$: magnetization density of any magnetic material.

Maxwell's Equations- Cont.

Gauss' Law - electrostatics

$$\nabla \cdot \mathbf{D} = 4\pi\rho$$

Gauss' Law - magnetostatics

$$\nabla \cdot \mathbf{B} = 0$$

Faraday's Law

$$\nabla \times \mathbf{E} = -\frac{1}{c}\dot{\mathbf{B}}$$

Ampere-Maxwell law

$$\nabla \times \mathbf{H} = \frac{4\pi}{c}\mathbf{J} + \frac{1}{c}\dot{\mathbf{D}}$$

Continuity Equation (Conservation of Charge)

$$\nabla \cdot \mathbf{J} + \dot{\rho} = 0$$

Comparing with **fluid dynamics**:

$$\nabla \cdot (\rho\mathbf{v}) + \dot{\rho} = 0$$

Energy Conservation + Poynting Vector

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}$$

with the **energy density** (erg cm⁻³) of an electromagnetic field

$$u = \frac{1}{8\pi} (\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H})$$

and the **Poynting vector or radiation energy flux** (erg cm⁻² s⁻¹)

$$\mathbf{S} = \frac{c}{4\pi} \mathbf{E} \times \mathbf{H}$$

The time variation of the energy density u consists of two parts: a spatial change of the energy flux $\mathbf{S}$ and a conversion of electromagnetic energy into **thermal energy** (**mechanical energy**).

For comparison, we have in **fluid dynamic**:

$$\frac{\partial e}{\partial t} + \nabla \cdot (e\mathbf{v}) = -p\nabla \cdot \mathbf{v}$$

Simple Example - Receiving antenna

Consider a straight wire of circular cross section carrying a steady current I . If all conditions are constant then $\dot{u} = 0$. Also think of the wire as a receiving antenna!

Here, $\mathbf{J} = \frac{I}{\pi R^2}$ and $\mathbf{E} = \frac{\mathbf{J}}{\sigma}$. So the thermal loss rate is

$$\frac{dW}{dl} = |\mathbf{E}|I$$

From Ampere's law, and integration over area:

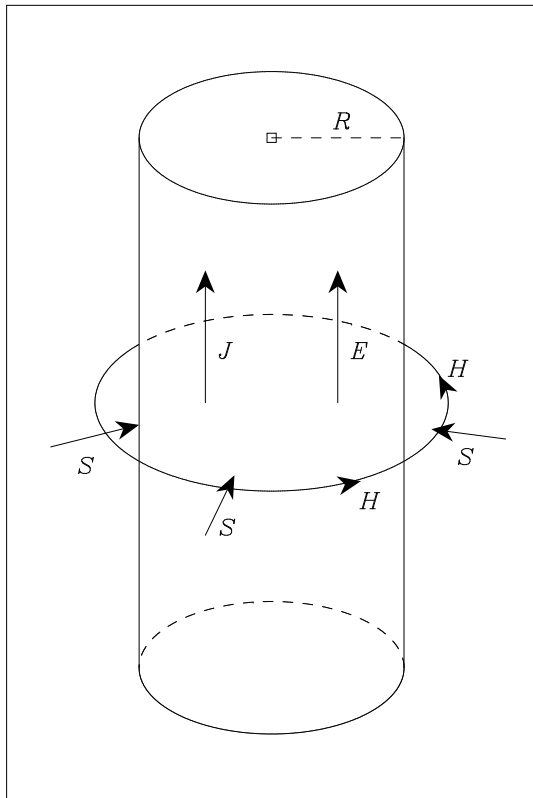
$$2\pi R \cdot \mathbf{H} = \frac{4\pi}{c} \mathbf{J} \cdot \pi R^2 \rightarrow |\mathbf{H}| = \frac{2I}{cR}$$

with a direction $\perp$ to $\mathbf{J}$. Then

$$|\mathbf{S}| = \frac{c}{4\pi} |\mathbf{E} \times \mathbf{H}| = \frac{|\mathbf{E}|I}{2\pi R}$$

Thus,

$$|\mathbf{E}|I = 2\pi R |\mathbf{S}| = \text{total flux of } \mathbf{S} \text{ at the surface of the wire}$$



The Wave Equation

Maxwell's eqs give the connection betw. the spatial and the time variation of the EM wave. Assuming **constant** σ , ϵ , and μ , we have (after some manipulation)

$$\nabla \times (\nabla \times \mathbf{H}) = \frac{1}{c} (4\pi\sigma + \epsilon \frac{\partial}{\partial t}) \nabla \times \mathbf{E} = -\frac{\mu}{c^2} (4\pi\sigma \dot{\mathbf{H}} + \epsilon \ddot{\mathbf{H}})$$

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{1}{c} (\nabla \times \mathbf{B}) = -\frac{\mu}{c^2} (4\pi\sigma \dot{\mathbf{E}} + \epsilon \ddot{\mathbf{E}})$$

Vector identity

$$\nabla \times (\nabla \times \mathbf{P}) = \nabla(\nabla \cdot \mathbf{P}) - \nabla^2 \mathbf{P}$$

No magnetic pole, $\therefore \nabla \cdot \mathbf{H} = 0$. Assuming $\rho = 0$, $\therefore \nabla \cdot \mathbf{E} = 0$. Thus,

$$\begin{aligned} \nabla^2 \mathbf{H} &= \frac{\epsilon\mu}{c^2} \ddot{\mathbf{H}} + \frac{4\pi\sigma\mu}{c^2} \dot{\mathbf{H}} \\ \nabla^2 \mathbf{E} &= \frac{\epsilon\mu}{c^2} \ddot{\mathbf{E}} + \frac{4\pi\sigma\mu}{c^2} \dot{\mathbf{E}} \end{aligned}$$

$\mathbf{E}$ and $\mathbf{H}$ are decoupled and both obey the same inhomogeneous wave eq.

Plane Wave in Nonconducting medium

Nonconducting (i.e., $\sigma = 0$), $\Rightarrow$ no currents no charges. Then $\mathbf{E}$ and $\mathbf{H}$ obey the homogeneous wave eq.:

$$\begin{aligned}\nabla^2 \mathbf{H} &= \frac{\epsilon\mu}{c^2} \ddot{\mathbf{H}} \rightarrow \nabla^2 \mathbf{H} - \frac{1}{v^2} \ddot{\mathbf{H}} = 0 \\ \nabla^2 \mathbf{E} &= \frac{\epsilon\mu}{c^2} \ddot{\mathbf{E}} \rightarrow \nabla^2 \mathbf{E} - \frac{1}{v^2} \ddot{\mathbf{E}} = 0\end{aligned}$$

For a simple case, each component of $\mathbf{E}$ and $\mathbf{H}$ will have a plane wave soln with $v = \frac{c}{\sqrt{\epsilon\mu}}$. Thus, a vector form of the wave moving in $\mathbf{k}$ direction is (in rectangular coors):

$$\mathbf{A}(\mathbf{x}, t) = \mathbf{A}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

with $\mathbf{A}_0$, $\mathbf{k}$, $\omega = \text{const.}$ This eq. has the following properties:

$$\begin{aligned}\dot{\mathbf{A}} &= -i\omega \mathbf{A} \\ \ddot{\mathbf{A}} &= -\omega^2 \mathbf{A} \\ \nabla \cdot \mathbf{A} &= i\mathbf{k} \cdot \mathbf{A} \\ \nabla^2 \mathbf{A} &= -k^2 \mathbf{A}\end{aligned}$$

Plane Wave Traveling in z Direction

Aligning the wave direction with z -axis ($\Rightarrow$ scalar), then (with A being either E or B)

$$\nabla^2 A - \frac{1}{v^2} \ddot{A} = 0$$

where

$$v = \frac{c}{\sqrt{\epsilon\mu}}$$

The solution is

$$A = A_0 e^{i(kz \pm \omega t)}$$

with

$$k^2 = \frac{\epsilon\mu}{c^2} \omega^2$$

The phase of the wave is $\varphi = kx \pm \omega t$. Points of constant phase travel with the phase velocity

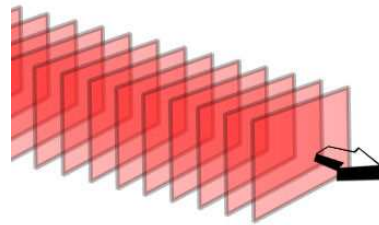
$$v = \frac{\omega}{k} = \frac{c}{\sqrt{\epsilon\mu}}$$

Thus, the index of refraction is

$$n = \frac{c}{v} = \sqrt{\epsilon\mu} = \frac{c}{\omega} k$$

Plane Wave Traveling in z Direction

A wave is plane if the surfaces of constant phase form planes $z = \text{const}$, ie., all components of the $\mathbf{E}$ and $\mathbf{H}$ field are independent of x and y for fixed z .



$\therefore \nabla \cdot \mathbf{H} = 0$ and $\nabla \cdot \mathbf{E} = 0$ (assuming $\rho = 0$), we also have

$$\frac{\partial E_z}{\partial z} = 0 \quad \text{and} \quad \frac{\partial H_z}{\partial z} = 0$$

From Faraday's and Ampere-Maxwell laws, we also have

$$\frac{\partial E_z}{\partial t} = 0 \quad \text{and} \quad \frac{\partial H_z}{\partial t} = 0$$

Thus E_z and H_z must be constant in space and time and are set to

$$E_z \equiv 0 \quad \text{and} \quad H_z = 0$$

The plane EM wave in a nonconducting medium is transverse, with $\mathbf{E}$ and $\mathbf{H} \perp z$.

Plane Wave - cont.

The only components which differ from zero are

$$\begin{aligned}\frac{\partial E_x}{\partial z} &= -\frac{\mu}{c} \frac{\partial H_y}{\partial t}, & \frac{\partial H_x}{\partial z} &= \frac{\epsilon}{c} \frac{\partial E_y}{\partial t} \\ \frac{\partial E_y}{\partial z} &= \frac{\mu}{c} \frac{\partial H_x}{\partial t}, & \frac{\partial H_y}{\partial z} &= -\frac{\epsilon}{c} \frac{\partial E_x}{\partial t}\end{aligned}$$

Thus,

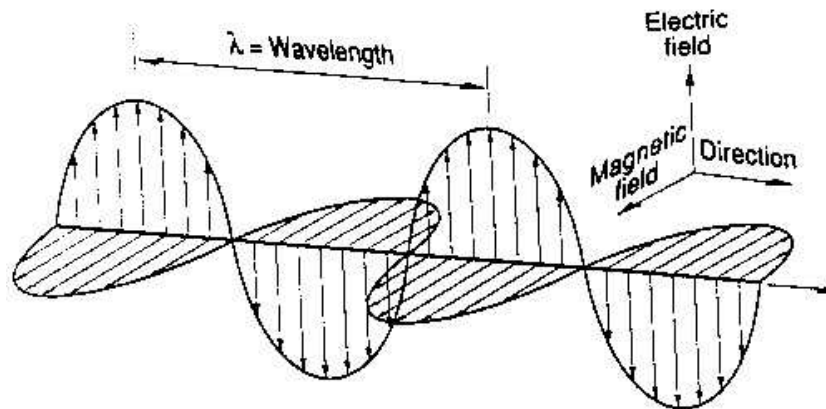
$$\begin{aligned}\frac{\partial E_x}{\partial z} &= ikE_x = -\frac{\mu}{c} \frac{\partial H_y}{\partial t} = \frac{i\omega\mu}{c} H_y \rightarrow H_y = \frac{ck}{\omega\mu} E_x \\ \frac{\partial E_y}{\partial z} &= ikE_y = \frac{\mu}{c} \frac{\partial H_x}{\partial t} = -\frac{i\omega\mu}{c} H_x \rightarrow H_x = -\frac{ck}{\omega\mu} E_y\end{aligned}$$

resulting in

$$\mathbf{E} \cdot \mathbf{H} = E_x H_x + E_y H_y = -\frac{ck}{\omega\mu} E_x E_y + \frac{ck}{\omega\mu} E_y E_x = 0$$

thus, $\mathbf{E} \perp \mathbf{H}$; together with the wave vector $\mathbf{k}$, these form an orthogonal system.

Plane Wave - cont.



Following the previous slide, the ratio of their absolute values are

$$\frac{|\mathbf{E}|}{|\mathbf{H}|} = \frac{\omega\mu}{ck} = \sqrt{\frac{\mu}{\epsilon}} \quad \therefore k = \frac{1}{\sqrt{\epsilon\mu}} \frac{\omega}{c}$$

The energy flux and the (time averaged) energy density of this wave is ($v = \frac{\omega}{k} = \frac{c}{\sqrt{\epsilon\mu}}$)

$$|\mathbf{S}| = \frac{c}{4\pi} \sqrt{\frac{\epsilon}{\mu}} \mathbf{E}^2 = \frac{\epsilon v}{4\pi} \mathbf{E}^2 = \frac{c}{4\pi} \sqrt{\frac{\mu}{\epsilon}} \mathbf{H}^2, \quad u = \frac{1}{8\pi} (\epsilon \mathbf{E} \cdot \mathbf{E}^* + \mu \mathbf{H} \cdot \mathbf{H}^*) = \frac{\epsilon}{4\pi} \mathbf{E}^2 = \frac{|\mathbf{S}|}{v}$$

and $\mathbf{S}$ points in the direction of the propagation vector $\mathbf{k}$.

Wave Packets and Group Velocity

A (1-D) wave of general shape can be formed by superposing simple harmonic waves

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k) e^{i(kx - \omega t)} dk$$

where $A(k)$ is the amplitude of the wave with the wave number k . The angular frequency of these waves is a function of k :

$$\omega = \omega(k)$$

this relation is referred to as the *dispersion relation* of the waves. If $A(k)$ is a fairly sharply peaked function around some k_0 , only waves with k not too different from k_0 will contribute, and a linear approximation for the dispersion relation will be sufficient:

$$\omega(k) \approx \omega_0 + \left. \frac{d\omega}{dk} \right|_0 (k - k_0) = \omega_0 - \left. \frac{d\omega}{dk} \right|_0 k_0 + \left. \frac{d\omega}{dk} \right|_0 k$$

Thus,

$$u(x, t) \approx \frac{1}{\sqrt{2\pi}} \exp[i(\left. \frac{d\omega}{dk} \right|_0 k_0 - \omega_0)t] \int_{-\infty}^{\infty} A(k) \exp[ik(x - \left. \frac{d\omega}{dk} \right|_0 t)] dk$$

Wave Packets - cont.

$$(1) \quad u(x, t) \approx \frac{1}{\sqrt{2\pi}} \exp\left[i\left(\frac{d\omega}{dk}\bigg|_0 k_0 - \omega_0\right)t\right] \int_{-\infty}^{\infty} A(k) \exp\left[ik\left(x - \frac{d\omega}{dk}\bigg|_0 t\right)\right] dk$$

At $t = 0$,

$$u(x, 0) \approx \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A(k) \exp[ikx] dk$$

Thus, Eq. 1 becomes

$$(2) \quad u(x, t) = u\left(x - \frac{d\omega}{dk}\bigg|_0 t, 0\right) \exp\left[i\left(\frac{d\omega}{dk}\bigg|_0 k_0 - \omega_0\right)t\right]$$

The exponential is only a phase factor. The wave packets travels undistorted in shape except for an overall phase factor with the group velocity

$$v_g = \frac{d\omega}{dk}$$

This is strictly true if ω is a linear function of k , so that $v_g = \text{const.}$ Otherwise, v_g will depend on k and the pulse will disperse. [Demo1 of wave packet](#) [Demo2 of wave packet](#)

Group Velocity

$$\omega = kv$$

Thus,

$$\frac{d\omega}{dk} = v_g = v + k \frac{dv}{dk} = v \left(1 + \frac{k}{v} \frac{dv}{dk} \right) = v \left(1 + \frac{\lambda}{n} \frac{dn}{d\lambda} \right)$$

where

$$n = \frac{c}{v}$$

$$\lambda = \frac{2\pi}{k}$$

Normally, we have a dispersion $dn/d\lambda < 0$, so that $v_g < v$. Energy and information are usually propagated with the group velocity.

Plane waves in Dissipative Media

In this case, $\sigma \neq 0$. Assuming strictly harmonic waves propagating in the direction of increasing x :

$$\mathbf{E}(x, t) = \mathbf{E}_0 e^{i(kx - \omega t)}$$

The wave eqs become

$$\left[k^2 - \left(\frac{\epsilon \mu}{c^2} \omega^2 + i \frac{4\pi \sigma \mu \omega}{c^2} \right) \right] \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix} = 0$$

To be valid for arbitrary $\mathbf{E}$ and $\mathbf{H}$, the square bracket must be zero, and the *dispersion relation* becomes (**Please remember it**)

$$k^2 = \frac{\mu \epsilon \omega^2}{c^2} \left(1 + i \frac{4\pi \sigma}{\omega \epsilon} \right)$$

Assuming σ is real and writing $k = a + ib$, then

$$\mathbf{E}(x, t) = \mathbf{E}_0 e^{-bx} e^{i(ax - \omega t)}$$

Thus, the conductivity (if real) gives rise to an exponential damping (absorption) of the wave.

Plane waves in Dissipative Media

Using the index of refraction $n = c/v$ and the absorption coefficient κ , then

$$\mathbf{E}(x, t) = \mathbf{E}_0 \exp\left(-\frac{\omega}{c} n \kappa x\right) \exp\left[i\omega\left(\frac{n}{c}x - t\right)\right]$$

with

$$n\kappa = \sqrt{\epsilon\mu} \sqrt{\frac{1}{2} \left(\sqrt{1 + \left(\frac{4\pi\sigma}{\epsilon\omega}\right)^2} - 1 \right)}$$
$$n = \sqrt{\epsilon\mu} \sqrt{\frac{1}{2} \left(\sqrt{1 + \left(\frac{4\pi\sigma}{\epsilon\omega}\right)^2} + 1 \right)}$$

Thus, the wave travels at v for a distance of $\frac{c}{n\kappa\omega} = \frac{1}{k\kappa}$

Dispersion Measure of a Tenuous Plasma

The simple model for dissipative medium is that of a tenuous plasma where free electrons and ions are uniformly distributed so that the total space charge density is zero. The free electrons are accelerated by the electric field intensity,

$$m_e \dot{\mathbf{v}} = m_e \ddot{\mathbf{r}} = -e \mathbf{E}_0 e^{-i\omega t}$$

with the soln,

$$\mathbf{v} = \frac{e}{im_e \omega} \mathbf{E}_0 e^{-i\omega t} = -i \frac{e}{m_e \omega} \mathbf{E}$$

Thus, the electron current density,

$$\mathbf{J} = -Ne\mathbf{v} = i \frac{Ne^2}{m_e \omega} \mathbf{E} = \sigma \mathbf{E}$$

Ions have current too, but smaller by a factor of $m_i/m_e \gtrsim 2 \times 10^3$ and can be ignored.

Dispersion Measure - Plasma Frequency

So the conductivity is purely imaginary, with

$$\sigma = i \frac{Ne^2}{m_e \omega}$$

The amplitude of the wave does not decrease with the distance. $\mathbf{J}$ and $\mathbf{E}$ are 90° out of phase with each other ($i = e^{i\pi/2}$). Thus there is no time-averaged mechanical work done on the particles by the field in an isotropic plasma, and no dissipation. So the wave either passes through the medium or is reflected, without energy loss.

For a thin medium with $\epsilon \approx 1$ and $\mu \approx 1$, then

$$k^2 = \frac{\omega^2}{c^2} \left(1 - \frac{\omega_p^2}{\omega^2}\right)$$

where the plasma frequency (a measure of the mobility of electron gas)

$$\omega_p^2 = \frac{4\pi Ne^2}{m_e}$$

$$\nu_p = \frac{\omega_p}{2\pi} = 8.97 \sqrt{\frac{N}{\text{cm}^{-3}}} \text{ kHz}$$

Plasma Frequency

$$k^2 = \frac{\omega^2}{c^2} \left(1 - \frac{\omega_p^2}{\omega^2}\right)$$

For $\omega > \omega_p$, k is real, the **phase velocity** is

$$v = \frac{\omega}{k} = \frac{c}{\sqrt{1 - \frac{\omega_p^2}{\omega^2}}} > c$$

the **group velocity** is

$$v_g = \frac{d\omega}{dk} = \frac{1}{dk/d\omega} = c \sqrt{1 - \frac{\omega_p^2}{\omega^2}} < c$$

which **increases with ω** . Thus,

$$vv_g = c^2$$

For $\omega = \omega_p$, $v_g = 0$, thus for waves with $\omega < \omega_p$ are reflected and no wave propagation is possible (remember the radio window observable from ground?)

Dispersion Measure: Radio pulsar

EM pulses are **dispersive in a plasma**. The arrival time of pulsar pulses depends on the frequency at which they are measured: the lower the observing freq., the later the pulse arrives. A pulse emitted by a pulsar at the distance L will be received at

$$\tau = \int_0^L \frac{dl}{v_g}$$

In the ISM, $N \sim 10^{-3} - 10^{-1} \text{ cm}^{-3}$ and $\nu_p \sim 2.85 - 0.285 \text{ kHz}$. However, the observing frequency must be $\nu > 10 \text{ MHz}$ in order to pass through the ionosphere. Thus, $\nu \gg \nu_p$ is needed and so

$$\frac{1}{v_g} = \frac{1}{c \sqrt{1 - \frac{\nu_p^2}{\nu^2}}} \approx \frac{1}{c} \left(1 + \frac{1}{2} \frac{\nu_p^2}{\nu^2} \right)$$

A pulse will be received at

$$\tau \approx \frac{1}{c} \int_0^L \left(1 + \frac{1}{2} \frac{\nu_p^2}{\nu^2} \right) dl = \frac{L}{c} + \frac{e^2}{2\pi c m_e} \frac{1}{\nu^2} \int_0^L N(l) dl$$

Dispersion Measure: Radio pulsar

$$\tau \approx \frac{L}{c} + \frac{e^2}{2\pi c m_e} \frac{\text{DM}}{\nu^2} \quad \text{with} \quad \text{DM} \equiv \int_0^L N(l) dl$$

The difference betw. the pulse arrival times measured at two freq., ν_1 and ν_2 , is then

$$\Delta\tau = \frac{e^2}{2\pi c m_e} \left[\frac{1}{\nu_1^2} - \frac{1}{\nu_2^2} \right] \text{DM}$$

Now, what is usually measured is the rate of change of arrival time w.r.t. freq.:

$$\frac{d\tau}{d\nu} = -\frac{e^2}{\pi c m_e \nu^3} \text{DM}$$

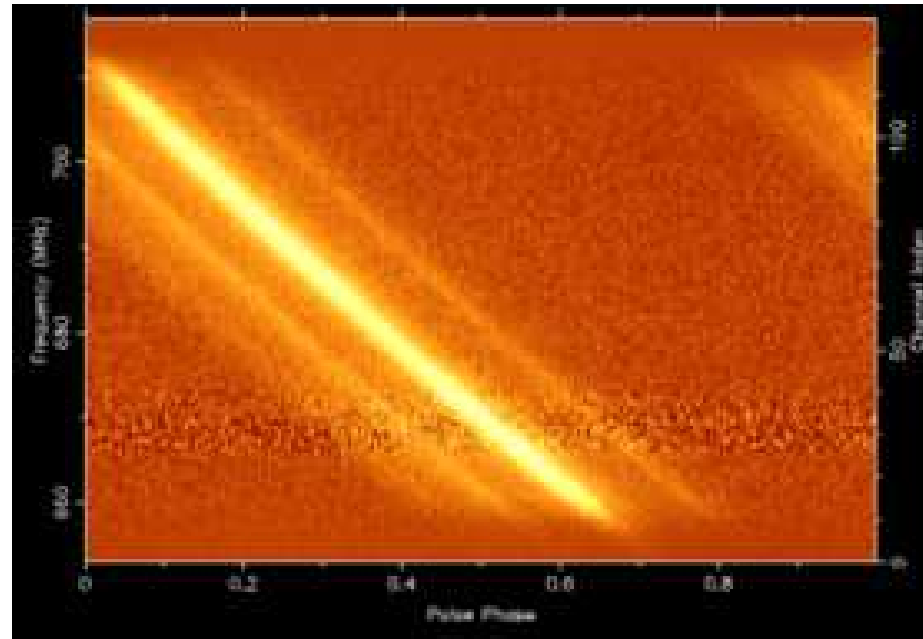
DM is usually measured in $\text{cm}^{-3} \text{ pc}$, with N in cm^{-3} and dl in pc. Therefore,

$$\frac{\Delta\tau}{\mu s} = 4.148 \times 10^3 \left[\frac{\text{DM}}{\text{cm}^{-3} \text{ pc}} \right] \left[\frac{1}{\left(\frac{\nu_1}{\text{MHz}}\right)^2} - \frac{1}{\left(\frac{\nu_2}{\text{MHz}}\right)^2} \right]$$

$$\frac{\text{DM}}{\text{cm}^{-3} \text{ pc}} = 2.410 \times 10^{-4} \frac{\Delta\tau}{\mu s} \left[\frac{1}{\left(\frac{\nu_1}{\text{MHz}}\right)^2} - \frac{1}{\left(\frac{\nu_2}{\text{MHz}}\right)^2} \right]^{-1}$$

Measuring $\Delta\tau$, we can derive DM, and then we can derive either the distance if the ISM N is known or the ISM N if the distance is known.

Dispersion Measure



The dispersed pulse profile of the 5.7 millisecond binary pulsar PSR J0437-4715. The observing radio frequency (~ 650 - 710 MHz with a mean of 680 MHz) is on the y axis and the pulse arrival time (μs) on the x axis. The observing frequency was swept from ~ 650 to 710 MHz. The dispersion measure can be determined by the slope of the SWEPT PULSE.

Dispersion Measure - Resolving a pulse

Dispersion ISM, combined with a finite pulse width, sets a limit to the fine structure one can resolve in a pulse. In terms of infinitesimal quantities, we have by differencing the original eq.,

$$\frac{\Delta\tau}{\mu s} = 4.148 \times 10^3 \left[\frac{\text{DM}}{\text{cm}^{-3} \text{ pc}} \right] \left[2 \frac{\Delta\nu}{\nu^3} \right]$$

i.e., let $\nu_1 = \nu$ and $\nu_2 = \nu + \Delta\nu$ with $\Delta\nu \ll \nu$, then the bracket in original eq. is $\approx 2 \frac{\Delta\nu}{\nu^3}$. Setting the bandwidth $b = \Delta\nu$ and time resolution $\tau_s = \Delta\tau$, we have

$$\frac{b}{\text{MHz}} = 1.205 \times 10^{-4} \frac{1}{\left[\frac{\text{DM}}{\text{cm}^{-3} \text{ pc}} \right]} \left[\frac{\nu}{\text{MHz}} \right]^3 \frac{\tau_s}{\mu s}$$

Since the pulses will have a finite width in both time and frequency, this eq. will give a limit to the max. bandwidth that can be used at a given freq. and DM if a time resolution τ_s is needed. E.g., $\tau_s = 0.1 \mu s$, $\nu = 100 \text{ MHz}$, $\text{DM} = 30 \text{ cm}^{-3} \text{ pc}$, the max. receiver bandwidth must be $< 400 \text{ kHz}$.